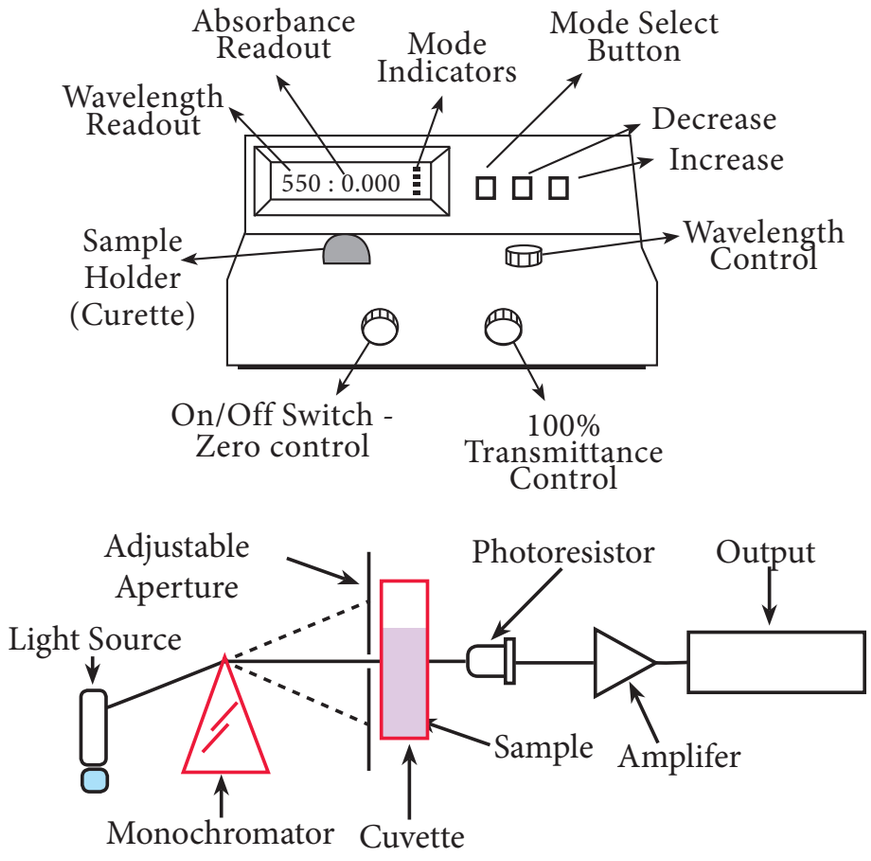

Figure 10.16 Colorimeter

In a colorimeter, beam of light is passed through an optical filter that transmits only a particular band of wavelength. The variation in the monochromatic light transmitted through a colorless sample (blank) and the amount of monochromatic light transmitted through a test sample is a measurement of the amount of monochromatic light absorbed by the sample. The monochromatic light absorbed is directly proportional to the concentration of the sample producing the colour and the path length through the sample.

Applications

A well known application of the colorimeter is to determine the concentration of the solute in a solution. To monitor the growth of a bacterial culture, colorimeter is used. As the culture grows, the medium becomes cloudier and absorbs more light which can be quantified. Colorimeters are used for testing water quality and screening of chemicals.

10.6.3 Colorimetric analysis

General steps in carrying out a photometric analysis is explained using blood glucose estimation procedure as an example.

a. Separation of the substance from the complex mixture. For example, estimation of blood glucose requires the precipitation of lipids and proteins using de-proteinising agents which otherwise interfere with the colour reaction of glucose.

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